Computer Architecture – Chapter-wise MCQs with Answers

Based on: Computer Architecture: A Quantitative Approach (Hennessy & Patterson, 6th Ed.)

Chapter 1: Fundamentals of Quantitative Design and Analysis

1. Execution time = ?
A) IC × CPI × Clock cycle time
B) CPI / Clock rate
C) IC × Clock rate
D) CPI × IC
Answer: A

2. Speedup is defined as
A) Told/Tnew
B) Tnew/Told
C) Perfnew/Perfold
D) Both A and C
Answer: D

3. Which law limits parallel speedup?
A) Moore's
B) Amdahl's
C) Dennard's
D) Murphy's
Answer: B

4. Which metric is best for user experience?
A) Throughput
B) CPI
C) Response time
D) Clock rate
Answer: C

5. Power $\propto$?
A) V
B) V^2f
C) f^2
D) C/V
Answer: B

6. Which improved performance most in 1990s?
A) Caches
B) ILP
C) Multicore
D) GPUs
Answer: B

7. Technology scaling mainly improved
A) Power only
B) Area only
C) Speed and density
D) Cost only
Answer: C

8. Throughput measures
A) Tasks/sec
B) Time/task
C) CPI
D) Frequency
Answer: A

9. Dependability includes
A) Reliability
B) Availability
C) Both
D) None
Answer: C

10. MTTF means
A) Mean time to failure
B) Max time to fail
C) Mean time to fix
D) None
Answer: A

11. Availability formula includes
A) MTTF
B) MTTR
C) Both
D) Neither
Answer: C

12. Weighted arithmetic mean is used when
A) Programs differ in importance
B) Same runtime
C) Same CPI
D) Same IC
Answer: A

13. Clock rate increase is limited by
A) Cache
B) Power
C) Memory
D) OS
Answer: B

14. Performance is inversely proportional to
A) CPI
B) Clock rate
C) Cores
D) Cache
Answer: A

15. Which is quantitative principle?
A) Make it faster
B) Make common case fast
C) Add cores
D) Increase cache
Answer: B

16. IC depends on
A) Hardware
B) Compiler
C) Program
D) All
Answer: D

17. Which era saw multicore rise?
A) 1980s
B) 1990s
C) 2004 onward
D) 1970s
Answer: C

18. Dennard scaling relates to
A) Voltage
B) Density
C) Power density
D) Cache
Answer: C

19. Cost per transistor trend is
A) Increasing
B) Constant
C) Decreasing
D) Random
Answer: C

20. Best metric for servers
A) Response time
B) Throughput
C) CPI
D) Clock
Answer: B

Chapter 2: Memory Hierarchy

1. Fastest memory
A) DRAM
B) SRAM
C) Flash
D) HDD
Answer: B

2. AMAT =
A) Hit + Miss
B) Hit + Miss rate × Miss penalty
C) Miss × Hit
D) Hit rate + Miss
Answer: B

3. Cache block contains
A) Single byte
B) Word
C) Multiple words
D) Page
Answer: C

4. Fully associative cache has
A) One set
B) Many sets
C) One line
D) No tags
Answer: A

5. Conflict misses occur in
A) Fully assoc
B) Direct mapped
C) Main memory
D) Registers
Answer: B

6. TLB stores
A) Instructions
B) Virtual to physical mappings
C) Cache blocks
D) Pages
Answer: B

7. Virtual memory benefit
A) Bigger memory
B) Protection
C) Sharing
D) All
Answer: D

8. Write-back updates
A) Cache only
B) Memory only
C) Both
D) Neither
Answer: A

9. Miss penalty mainly depends on
A) CPU
B) Next level memory
C) Cache
D) Compiler
Answer: B

10. Increasing block size increases
A) Capacity misses
B) Conflict misses
C) Compulsory misses
D) Hit rate always
Answer: C

11. Set associative reduces
A) Capacity misses
B) Conflict misses
C) Compulsory
D) All
Answer: B

12. DRAM is slower due to
A) Refresh
B) Capacitors
C) Sense amps
D) All
Answer: D

13. Page fault occurs when
A) Cache miss
B) TLB miss
C) Page not in memory
D) Disk error
Answer: C

14. Inclusive cache means
A) L1 contains L2
B) L2 contains L1
C) No sharing
D) Same size
Answer: B

15. Prefetching helps reduce
A) Hit time
B) Miss rate
C) Miss penalty
D) CPI
Answer: B

16. Victim cache reduces
A) Capacity misses
B) Conflict misses
C) Compulsory
D) All
Answer: B

17. Memory hierarchy exploits
A) Temporal locality
B) Spatial locality
C) Both
D) None
Answer: C

18. Page size affects
A) TLB miss
B) Disk access
C) Both
D) Neither
Answer: C

19. Which is non-volatile
A) SRAM
B) DRAM
C) Flash
D) Cache
Answer: C

20. Main memory technology
A) SRAM
B) DRAM
C) Flash
D) SSD
Answer: B

Chapter 3: Instruction-Level Parallelism

1. Pipeline improves
A) Latency
B) Throughput
C) CPI
D) IC
Answer: B

2. Data hazard occurs when
A) Same resource
B) Branch
C) Data dependency
D) Cache miss
Answer: C

3. RAW means
A) Read after write
B) Read after wait
C) Run after write
D) None
Answer: A

4. Tomasulo helps with
A) Branch
B) Data hazards
C) Cache
D) IO
Answer: B

5. Speculation is based on
A) Compiler
B) Branch prediction
C) Cache
D) Registers
Answer: B

6. Superscalar means
A) Multiple cores
B) Multiple issue
C) Multithread
D) SIMD
Answer: B

7. Register renaming removes
A) RAW
B) WAR
C) WAW
D) Both B and C
Answer: D

8. Branch predictor reduces
A) Miss rate
B) Penalty
C) Control hazard
D) Data hazard
Answer: C

9. Scoreboarding tracks
A) Cache
B) Hazards
C) Threads
D) IO
Answer: B

10. ILP mainly exploited by
A) Hardware
B) OS
C) User
D) Network
Answer: A

11. Out-of-order execution improves
A) IC
B) CPI
C) Cache
D) Power
Answer: B

12. Reservation stations store
A) Data
B) Instructions waiting
C) Cache
D) Threads
Answer: B

13. Speculative execution may cause
A) Correct results
B) Wasted work
C) Security issues
D) All
Answer: D

14. Commit stage ensures
A) Correct order
B) Fast exec
C) Cache update
D) Branch
Answer: A

15. Multiple issue increases
A) IC
B) IPC
C) Latency
D) Miss rate
Answer: B

16. Dynamic scheduling done by
A) Compiler
B) Hardware
C) OS
D) User
Answer: B

17. Branch target buffer stores
A) Data
B) Targets
C) Registers
D) Cache
Answer: B

18. Long pipelines increase
A) Miss penalty
B) Branch penalty
C) IC
D) Cache
Answer: B

19. Control hazards due to
A) Memory
B) Branches
C) Data
D) IO
Answer: B

20. CPI ideal pipeline is
A) 0
B) 0.5
C) 1
D) >1
Answer: C

Chapter 4: Data-Level Parallelism and GPUs

1. Vector processors operate on
A) Scalars
B) Arrays
C) Objects
D) Files
Answer: B

2. SIMD means
A) Single instruction multiple data
B) Same instruction many data
C) Both
D) None
Answer: A

3. GPU optimized for
A) Latency
B) Throughput
C) IO
D) Branch
Answer: B

4. Warp is
A) Thread block
B) Group of threads
C) Cache
D) Register
Answer: B

5. Occupancy means
A) Cache use
B) Active warps
C) Memory use
D) Power
Answer: B

6. Roofline model uses
A) CPI
B) Bandwidth
C) Intensity
D) Both B and C
Answer: D

7. Vector length register helps
A) Flexibility
B) Cache
C) Power
D) Branch
Answer: A

8. SIMT used in
A) CPUs
B) GPUs
C) DRAM
D) SSD
Answer: B

9. Memory coalescing improves
A) Latency
B) Bandwidth
C) Power
D) Cache
Answer: B

10. GPU cores are called
A) ALUs
B) SMs
C) CPUs
D) TPUs
Answer: B

11. DLP best for
A) Databases
B) Graphics
C) IO
D) OS
Answer: B

12. Vector masking used for
A) Control flow
B) Errors
C) Cache
D) Paging
Answer: A

13. Shared memory in GPU is
A) Slow
B) Global
C) On-chip
D) Disk
Answer: C

14. Arithmetic intensity is
A) FLOPs/byte
B) CPI
C) Cache size
D) Clock
Answer: A

15. GPU hides latency using
A) Caches
B) Multithreading
C) Branch pred
D) IO
Answer: B

16. SIMD width refers to
A) Registers
B) Lanes
C) Cores
D) Caches
Answer: B

17. Divergence reduces
A) Accuracy
B) Performance
C) Power
D) Memory
Answer: B

18. Vector pipeline differs by
A) Length
B) Chaining
C) Memory
D) Cache
Answer: B

19. GPU global memory is
A) Fastest
B) On-chip
C) DRAM
D) Register
Answer: C

20. Best workloads for GPU
A) Serial
B) Irregular
C) Data-parallel
D) OS
Answer: C

Chapter 5: Thread-Level Parallelism

1. TLP exploits
A) Instructions
B) Threads
C) Data
D) Cache
Answer: B

2. SMP means
A) Single memory processor
B) Shared memory processors
C) Serial machines
D) SIMD
Answer: B

3. Cache coherence ensures
A) Same value seen
B) Fast cache
C) Low power
D) No misses
Answer: A

4. MESI protocol states include
A) Modified
B) Exclusive
C) Shared
D) All
Answer: D

5. Directory protocol used in
A) Small SMP
B) Large systems
C) GPUs
D) PCs
Answer: B

6. False sharing occurs when
A) Same block diff vars
B) Same var
C) Diff caches
D) Disk
Answer: A

7. Barrier is
A) Lock
B) Sync point
C) Cache
D) IO
Answer: B

8. Spinlock wastes
A) Memory
B) CPU cycles
C) Disk
D) Cache
Answer: B

9. Consistency model defines
A) Cache
B) Order of memory ops
C) Power
D) IO
Answer: B

10. Sequential consistency is
A) Weak
B) Strong
C) Random
D) GPU
Answer: B

11. NUMA means
A) Uniform memory
B) Non-uniform memory
C) No memory
D) GPU memory
Answer: B

12. Scalability limited by
A) Cache
B) Sync
C) Amdahl
D) All
Answer: D

13. Atomic operation ensures
A) Speed
B) Mutual exclusion
C) Cache
D) IO
Answer: B

14. Lock-free uses
A) No sync
B) Atomic ops
C) Cache
D) Disk
Answer: B

15. Coherence traffic increases with
A) Cores
B) Cache size
C) Disk
D) IO
Answer: A

16. Write invalidate means
A) Others invalid
B) Write through
C) Cache flush
D) Disk write
Answer: A

17. Thread scheduling done by
A) Compiler
B) OS
C) Cache
D) GPU
Answer: B

18. Load imbalance reduces
A) Accuracy
B) Speedup
C) Power
D) Cache
Answer: B

19. Critical section protected by
A) Semaphore
B) Lock
C) Cache
D) TLB
Answer: B

20. Best TLP for
A) Serial
B) Parallel apps
C) IO
D) OS only
Answer: B

Chapter 6: Warehouse-Scale Computers

1. WSC designed for
A) Single user
B) Many users
C) Embedded
D) GPU only
Answer: B

2. MapReduce supports
A) ILP
B) DLP
C) RLP
D) TLP
Answer: C

3. WSC focus metric
A) CPI
B) Throughput/$
C) Clock
D) Cache
Answer: B

4. Energy efficiency critical due to
A) Cost
B) Cooling
C) Power limit
D) All
Answer: D

5. Data centers use
A) HDD only
B) SSD only
C) Both
D) Tape
Answer: C

6. Virtualization enables
A) Isolation
B) Sharing
C) Migration
D) All
Answer: D

7. Load balancer distributes
A) Power
B) Requests
C) Memory
D) Disk
Answer: B

8. Fault tolerance achieved by
A) Replication
B) Checkpoint
C) Restart
D) All
Answer: D

9. Network topology affects
A) Latency
B) Bandwidth
C) Cost
D) All
Answer: D

10. PUE measures
A) CPU power
B) Total/IT power
C) Cache
D) Disk
Answer: B

11. WSC storage often uses
A) RAID
B) Replication
C) Both
D) None
Answer: C

12. Elasticity means
A) Fixed size
B) Scale up/down
C) Fast CPU
D) Big cache
Answer: B

13. Microservices mean
A) Small services
B) Big servers
C) OS
D) GPU
Answer: A

14. Cold storage uses
A) SSD
B) HDD
C) Tape
D) DRAM
Answer: C

15. Latency-sensitive apps need
A) Fast response
B) High throughput
C) Big cache
D) GPU
Answer: A

16. Racks contain
A) CPUs
B) Servers
C) Switches
D) All
Answer: D

17. Software handles failures because
A) Cheap hardware
B) Many failures
C) Scale
D) All
Answer: D

18. WSC workload example
A) Web search
B) Gaming
C) OS boot
D) Compiler
Answer: A

19. Disaggregation means
A) Combine
B) Separate resources
C) Cache
D) GPU
Answer: B

20. Cloud pricing is based on
A) Time
B) Resources used
C) Both
D) None
Answer: C

Chapter 7: Domain-Specific Architectures

1. DSA optimized for
A) All tasks
B) One domain
C) OS
D) IO
Answer: B

2. TPU accelerates
A) Graphics
B) ML
C) Databases
D) OS
Answer: B

3. DSA benefit is
A) Speed
B) Energy
C) Area
D) All
Answer: D

4. Reduced precision helps
A) Power
B) Speed
C) Storage
D) All
Answer: D

5. Accelerators offload
A) OS
B) Specific tasks
C) Cache
D) IO
Answer: B

6. FPGA advantage
A) Fixed
B) Reconfigurable
C) Slow
D) Cheap
Answer: B

7. TPU uses
A) SIMD
B) Systolic arrays
C) Caches
D) CPUs
Answer: B

8. DSA trade-off is
A) Flexibility
B) Cost
C) Programmability
D) All
Answer: D

9. Inference differs from training by
A) Precision
B) Workload
C) Both
D) None
Answer: C

10. Edge devices need
A) Low power
B) High speed
C) Big memory
D) Disk
Answer: A

11. DSA design guided by
A) Workload
B) Compiler
C) Cache
D) OS
Answer: A

12. ASIC means
A) General CPU
B) Fixed hardware
C) FPGA
D) GPU
Answer: B

13. Memory bandwidth critical for
A) ML
B) IO
C) OS
D) Boot
Answer: A

14. Co-design means
A) HW+SW together
B) Cache+CPU
C) Disk+IO
D) None
Answer: A

15. Tensor cores used for
A) Graphics
B) Matrix ops
C) IO
D) Cache
Answer: B

16. DSA reduces
A) Generality
B) Cost per op
C) Energy per op
D) Both B and C
Answer: D

17. Training accelerators need
A) High precision
B) High bandwidth
C) Both
D) None
Answer: C

18. Inference accelerators prioritize
A) Accuracy
B) Latency
C) Cost
D) Disk
Answer: B

19. Heterogeneous systems include
A) CPU only
B) GPU only
C) CPU+accelerators
D) Disk
Answer: C

20. Future trend is
A) Single core
B) General CPU only
C) More DSAs
D) Slower tech
Answer: C